16 Here are the first five terms of a quadratic sequence.

10 21 38 61 90

Find an expression, in terms of n, for the nth term of this sequence.

First differences: 11, 17, 23, 29; second differences: 6.

So quadratic term is $3n^2$. Subtract it: 7, 9, 11, 13, 15.

Remainder is 2n + 5; nth term = $3n^2 + 2n + 5$.

3 marks total
Second diff 6 gives $3n^2$
Linear remainder 2n+5
$3n^2+2n+5$

.....

(Total for Question 16 is 3 marks)

17 Write down the coordinates of the turning point on the graph of $y = (x + 12)^2 - 7$

(..... , ...)

(Total for Question 17 is 1 mark)

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